

Simulation of Snowflake Growth

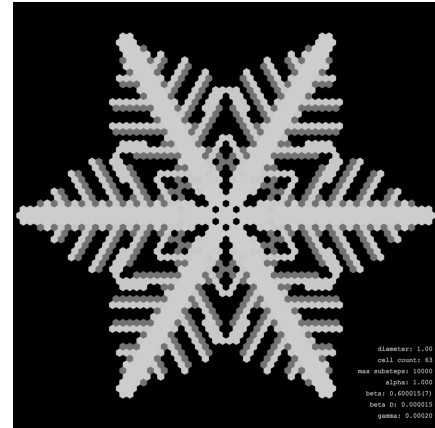
Simulation Developer / Dec 2023 – Jan 2024

Overview:

This project involved implementing an established method for simulating snowflake crystal growth, aiming to model realistic structural patterns influenced by environmental variables. It utilized a hexagonal cell grid and diffusion-based growth with threshold crystallization, guided by time-varying conditions such as temperature and humidity. While the approach produced interesting results and valuable insights, limitations in realism and performance were noted due to the technique's assumptions and the toolset used.

Tools & Skills:

Blender (Geometry Nodes), Simulation Design, Computational Optimization, Environmental Modeling, Data Visualization, Adapting and Extending Existing Methods



Example showing dendritic growth.

Objective:

Implement an existing snowflake crystal growth technique to simulate realistic structural patterns influenced by environmental conditions.

Design Highlights:

- Employed a hexagonal cell grid with symmetrical growth rules to optimize spatial efficiency and computational performance.
- Simulated diffusion-based growth by averaging water content across the grid, excluding areas near the growing crystal.
- Implemented a threshold-based crystallization model, designating cells as ice once local water concentration exceeded a defined limit.
- Used time-varying environmental variables such as humidity and temperature to guide and constrain crystal development.
- Created stage-like growth behavior by modulating environmental conditions across simulation steps.

Reflections:

The simulation generated interesting asymmetrical patterns, particularly with random initial conditions, but performance declined notably as grid size increased to match symmetrical models. The approach's realism was limited because the technique artificially increased mass near crystals, resulting in unrealistic accumulation. Blender Geometry Nodes lacked the optimization necessary for efficient large-scale simulation, suggesting that Python or MATLAB might better support future iterations. Despite these challenges, the project produced usable results and valuable insights.

References:

Gravner J., Griffeath D. *Modeling snow crystal growth II*. Physica D Nonlinear Phenomena, 2008. <https://doi.org/10.1016/j.physd.2007.09.008>.

Schwind M. (Entagma). *Creating A Procedural Snowflake*. YouTube, 2019. <https://www.youtube.com/watch?v=n4AQ3yj3L4w>